

Pat Davis

Embedded Systems & IoT Firmware Engineer • 7 Years Experience • MERIQ Index: 79% (Strong Candidate)

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TECHNICAL PROFICIENCIES & CERTIFICATIONS

Core Technologies: Embedded Linux, PCB Debugging, I2C/SPI/UART, C/C++, Python, ARM Cortex, FreeRTOS

Verified Certification: Arm Accredited Engineer (AAE)

PROFESSIONAL WORK EXPERIENCE

Senior Embedded Systems & IoT Firmware Engineer — *Tesla* (2023 - Present (3 yrs))

- Spearheaded architectural redesign of core Embedded Linux service, reducing latency by 54%.
- Led a squad of 6 engineers delivering scalable microservices processing 31M requests/day.
- Implemented automated testing & CI/CD pipelines increasing deployment frequency by 36%.

Associate Embedded Systems & IoT Firmware Engineer — *Bosch* (2019 - 2023)

- Designed and maintained PCB Debugging data endpoints with 97% SLA uptime.
- Optimized database indexing and caching layers using I2C/SPI/UART saving \$37k/yr cloud costs.

NOTABLE TECHNICAL PROJECTS

Automotive CAN-Bus Telemetry Gateway with Embedded Linux and Secure Boot [Embedded Linux, PCB Debugging, I2C/SPI/UART, C/C++]

Engineered end-to-end production solution leveraging Embedded Linux and PCB Debugging with full test automation.

Smart Grid Power Monitoring Device with MQTT & Hardware Cryptographic Accelerator [I2C/SPI/UART, C/C++, Python, ARM Cortex]

Architected high-resilience system integrating I2C/SPI/UART with 99.9% uptime.

EDUCATION & CREDENTIALS

B.E. in Information Technology

University of Toronto — 2015 - 2019 (Score: 8.81 / 10.0)